REMS - RAD comparison

Closest available timestamps and environmental readings | 26-27 September 2019 | Prepared 6 November 2026

88 RAD dose observations compared with 31,129 REMS samples. The following tables select the closest available REMS sample to each RAD UTC start, using estimated REMS UTC. They are a new nearest-time comparison, distinct from the earlier header-LMST interval join.

Observation 00: timing check	RAD	REMS / derived comparison
UTC at RAD start	2019-09-26 08:25:42	Same target instant
Mars clock at that instant	2538 00:15:58 (header)	2538 00:02:51 (estimated LMST)
Clock discrepancy	RAD header ahead	About 13 min 7 sec in Mars-clock units
Dose rate B / E	11.59364 / 12.77570 microGy/h	Not a REMS measurement

Two differences are shown separately. "UTC gap" is nearest REMS sample time minus RAD start, in Earth seconds. "Clock delta" is RAD header LMST minus REMS LMST estimated at the *same RAD UTC instant*, in Mars minutes:seconds. The 13:07 value is a clock discrepancy, not the distance to the nearest sample, and is not assumed constant.

Method and limits. REMS UTC is estimated by linear interpolation between the label START_TIME / STOP_TIME and the first / last table LMST. This is a diagnostic approximation, not a validated SPICE clock conversion. Nearest samples can lie before the observation or across data gaps; a nearest row does not establish simultaneous coverage. No timing offset has been applied.

Reading the tables. B/E are observation-average dose rates; pressure and ambient temperature are individual nearest-sample readings. "Window N" retains the earlier provisional header-LMST interval count. That join found 44 observations with samples, 42 with zero RAD QC. Environmental values have not been filtered by sensor flags.

Sources. Supplied RAD TXT/LBL (RAD_RDR_2019_269_08_25_2538_V00); REMS TAB/LBL (RME_622755917RMD25380000000_____P2); MODRDR6.FMT. RAD RDR SIS section 4.2.1.2 describes the Mars header as LMST: PDS instrument documentation. All numeric comparisons are derived from the supplied files.

Closest-time comparison | 1 of 5

RAD observations 00 to 34 | Sol 2538 | REMS times rounded to nearest second for display

RAD obs.	RAD UTC 2019	RAD Mars HH:MM:SS	Nearest REMS UTC est. 2019	REMS LMST HH:MM:SS	UTC gap Earth s	Clock delta Mars m:s	RAD B / E microGy/h	REMS P Pa	REMS air K	REMS flags P / air	Window N
00	09-26 08:25:42	0:15:58	09-26 08:25:42	00:02:51	+0.3	+13:07	11.59364 12.77570	857.81	200.61	11 011XX1111	961
02	09-26 08:42:48	0:32:37	09-26 08:42:48	00:19:30	+0.3	+13:07	12.35597 12.70742	858.69	200.75	11 011XX1111	961
04	09-26 08:59:54	0:49:16	09-26 08:59:54	00:36:08	+0.3	+13:08	11.60890 12.92936	859.85	199.82	11 011XX1111	961
06	09-26 09:17:01	1:05:55	09-26 09:17:01	00:52:48	+0.3	+13:07	11.60840 12.67108	860.91	199.41	11 011XX1111	12
08	09-26 09:34:07	1:22:34	09-26 09:30:41	01:06:06	-205.7	+13:08	11.97723 12.91626	861.81	200.08	11 011XX1111	0
10	09-26 09:51:13	1:39:13	09-26 09:30:41	01:06:06	-1231.7	+13:08	11.27921 12.68726	861.81	200.08	11 011XX1111	0
12*	09-26 10:08:15	1:55:47	09-26 10:27:19	02:01:13	+1144.3	+13:08	10.60040 11.72766	864.67	198.29	11 011XX1111	308
14	09-26 10:23:20	2:10:28	09-26 10:27:19	02:01:13	+239.3	+13:08	11.21340 12.70516	864.67	198.29	11 011XX1111	0
16	09-26 10:40:25	2:27:06	09-26 10:32:26	02:06:12	-478.7	+13:08	11.30194 12.36053	865.05	199.13	11 011XX1111	0
18	09-26 10:57:30	2:43:44	09-26 10:32:26	02:06:12	-1503.7	+13:09	11.04656 12.96713	865.05	199.13	11 011XX1111	0
20	09-26 11:14:36	3:00:22	09-26 11:29:08	03:01:23	+872.4	+13:08	12.00939 12.64079	864.84	196.74	11 011XX1101	308
22	09-26 11:31:42	3:17:01	09-26 11:31:42	03:03:53	+0.4	+13:09	13.24353 12.66908	864.64	196.37	11 011XX1111	0
24	09-26 11:48:48	3:33:40	09-26 11:34:15	03:06:22	-872.6	+13:09	12.06343 12.74120	864.77	197.10	11 011XX1101	0
26	09-26 12:05:54	3:50:18	09-26 12:30:44	04:01:20	+1490.4	+13:08	12.21572 12.89983	860.72	193.02	11 011XX1101	281
28	09-26 12:23:01	4:06:58	09-26 12:30:44	04:01:20	+463.4	+13:09	11.18590 12.79784	860.72	193.02	11 011XX1101	0
30	09-26 12:40:07	4:23:37	09-26 12:35:51	04:06:19	-255.6	+13:09	11.10157 12.60096	860.06	193.59	11 011XX1111	0
32	09-26 12:57:13	4:40:15	09-26 12:35:51	04:06:19	-1281.6	+13:09	11.24528 12.53969	860.06	193.59	11 011XX1111	0
34	09-26 13:14:19	4:56:54	09-26 13:32:24	05:01:21	+1085.4	+13:09	12.49338 12.71520	858.14	194.14	11 011XX1111	687

* RAD QC = 00000001; otherwise 00000000. [R] = inferred rollover to sol 2539 (original header: 2538). REMS sol = 2538. -- = missing (-999).

P / air = raw sensor confidence flags; zeros can denote instrument state. No maximum UTC gap was imposed. Clock delta is evaluated at RAD UTC.

Closest-time comparison | 2 of 5

RAD observations 36 to 70 | Sol 2538 | REMS times rounded to nearest second for display

RAD obs.	RAD UTC 2019	RAD Mars HH:MM:SS	Nearest REMS UTC est. 2019	REMS LMST HH:MM:SS	UTC gap Earth s	Clock delta Mars m:s	RAD B / E microGy/h	REMS P Pa	REMS air K	REMS flags P / air	Window N
36	09-26 13:31:25	5:13:33	09-26 13:32:24	05:01:21	+59.4	+13:10	12.04744 12.44899	858.14	194.14	11 011XX1111	961
38	09-26 13:48:31	5:30:11	09-26 13:48:31	05:17:02	+0.4	+13:09	11.70327 12.66115	858.41	193.01	11 011XX1111	961
40	09-26 14:05:37	5:46:50	09-26 14:05:37	05:33:41	+0.4	+13:10	11.19332 12.49547	859.07	195.29	11 011XX0011	961
42	09-26 14:22:43	6:03:29	09-26 14:22:43	05:50:19	+0.5	+13:10	11.81159 12.63499	860.99	192.70	11 011XX0011	961
44	09-26 14:39:49	6:20:07	09-26 14:39:49	06:06:58	+0.5	+13:10	10.79328 12.75619	862.31	192.43	11 011XX0011	961
46	09-26 14:56:55	6:36:46	09-26 14:56:55	06:23:36	+0.5	+13:10	11.01427 12.76374	864.38	194.59	11 011XX0001	961
48	09-26 15:14:01	6:53:25	09-26 15:14:01	06:40:15	+0.5	+13:10	11.38620 12.58650	866.00	201.92	11 011XX0001	797
50	09-26 15:31:07	7:10:03	09-26 15:31:07	06:56:54	+0.5	+13:10	11.75178 12.57236	867.47	202.28	11 011XX0011	0
52	09-26 15:48:13	7:26:42	09-26 15:41:09	07:06:39	-423.5	+13:10	11.81130 12.70480	868.14	206.86	11 011XX0011	0
54	09-26 16:05:19	7:43:21	09-26 15:41:09	07:06:39	-1449.5	+13:11	11.35112 12.70657	868.14	206.86	11 011XX0011	0
56	09-26 16:22:25	8:00:00	09-26 16:37:06	08:01:06	+880.5	+13:11	11.36688 12.51932	865.80	205.65	11 111XX0011	308
58	09-26 16:39:31	8:16:38	09-26 16:39:31	08:03:27	-0.5	+13:11	10.96666 12.54363	865.63	206.43	11 111XX0011	0
60	09-26 16:56:37	8:33:17	09-26 16:42:13	08:06:04	-864.5	+13:11	11.56116 12.67733	865.38	206.03	11 111XX0001	0
62	09-26 17:13:43	8:49:56	09-26 17:38:30	09:00:51	+1486.6	+13:12	11.94855 12.74024	855.84	213.10	11 111XX0101	288
64	09-26 17:30:24	9:06:10	09-26 17:38:30	09:00:51	+485.6	+13:11	11.22728 12.78598	855.84	213.10	11 111XX0101	0
66	09-26 17:47:29	9:22:48	09-26 17:43:37	09:05:50	-232.4	+13:12	12.10068 12.70856	855.12	217.07	11 111XX0111	0
68	09-26 18:04:34	9:39:25	09-26 17:43:37	09:05:50	-1257.4	+13:11	11.67557 12.73592	855.12	217.07	11 111XX0111	0
70	09-26 18:21:39	9:56:03	09-26 18:40:04	10:00:46	+1104.6	+13:12	11.23398 12.60605	846.83	210.61	11 111XX1111	308

* RAD QC = 00000001; otherwise 00000000. [R] = inferred rollover to sol 2539 (original header: 2538). REMS sol = 2538. -- = missing (-999).

P / air = raw sensor confidence flags; zeros can denote instrument state. No maximum UTC gap was imposed. Clock delta is evaluated at RAD UTC.

Closest-time comparison | 3 of 5

RAD observations 72 to 106 | Sol 2538 | REMS times rounded to nearest second for display

RAD obs.	RAD UTC 2019	RAD Mars HH:MM:SS	Nearest REMS UTC est. 2019	REMS LMST HH:MM:SS	UTC gap Earth s	Clock delta Mars m:s	RAD B / E microGy/h	REMS P Pa	REMS air K	REMS flags P / air	Window N
72	09-26 18:38:44	10:12:41	09-26 18:40:04	10:00:46	+79.6	+13:12	11.72574 12.61535	846.83	210.61	11 111XX1111	0
74	09-26 18:55:50	10:29:19	09-26 18:45:11	10:05:45	-639.4	+13:12	12.16544 12.59196	846.35	210.88	11 111XX1101	0
76	09-26 19:12:56	10:45:58	09-26 18:45:11	10:05:45	-1665.4	+13:12	11.41097 12.84512	846.35	210.88	11 111XX1101	38
78	09-26 19:30:02	11:02:37	09-26 19:41:54	11:00:57	+711.6	+13:12	11.59952 12.98967	838.63	214.97	10 111XX1101	961
80	09-26 19:47:08	11:19:15	09-26 19:47:08	11:06:03	-0.4	+13:12	12.27787 12.71261	837.96	218.92	11 111XX1111	961
82	09-26 20:04:14	11:35:54	09-26 20:04:14	11:22:41	-0.4	+13:12	11.55873 13.00927	836.34	220.94	11 111XX1111	961
84	09-26 20:21:20	11:52:33	09-26 20:21:20	11:39:20	-0.3	+13:13	11.82000 12.89431	834.47	219.33	11 111XX1101	946
86	09-26 20:38:26	12:09:11	09-26 20:38:26	11:55:58	-0.3	+13:12	11.81327 13.03618	832.49	221.13	11 111XX1111	961
88*	09-26 20:55:22	12:25:40	09-26 20:55:22	12:12:27	-0.3	+13:12	7.25048 8.15563	830.04	223.68	11 111XX1101	649
90	09-26 21:07:14	12:37:13	09-26 21:07:14	12:24:00	-0.3	+13:12	14.18208 15.08608	828.30	222.47	11 111XX1111	961
92	09-26 21:24:18	12:53:50	09-26 21:24:18	12:40:37	-0.3	+13:13	11.22720 13.02993	825.91	229.90	11 111XX1111	762
94	09-26 21:41:23	13:10:28	09-26 21:41:23	12:57:14	-0.3	+13:13	11.04075 13.33796	823.25	234.28	10 111XX0011	0
96	09-26 21:58:28	13:27:05	09-26 21:50:49	13:06:25	-459.3	+13:13	12.29756 12.96942	821.95	235.69	11 111XX0001	0
98	09-26 22:15:33	13:43:43	09-26 21:50:49	13:06:25	-1484.3	+13:13	11.58214 12.87804	821.95	235.69	11 111XX0001	0
100	09-26 22:32:38	14:00:21	09-26 22:46:44	14:00:51	+845.7	+13:14	12.62168 12.96079	816.49	231.86	10 111XX0011	308
102	09-26 22:49:42	14:16:58	09-26 22:49:42	14:03:44	-0.3	+13:14	11.48983 12.98200	816.41	236.64	10 111XX0001	0
104	09-26 23:06:48	14:33:36	09-26 22:51:51	14:05:49	-897.3	+13:13	11.37440 12.80608	816.31	233.37	11 111XX0001	0
106	09-26 23:23:54	14:50:15	09-26 23:48:18	15:00:46	+1463.8	+13:14	12.90837 13.23256	813.55	241.45	10 111XX0011	308

* RAD QC = 00000001; otherwise 00000000. [R] = inferred rollover to sol 2539 (original header: 2538). REMS sol = 2538. -- = missing (-999).

P / air = raw sensor confidence flags; zeros can denote instrument state. No maximum UTC gap was imposed. Clock delta is evaluated at RAD UTC.

Closest-time comparison | 4 of 5

RAD observations 108 to 142 | Sol 2538 | REMS times rounded to nearest second for display

RAD obs.	RAD UTC 2019	RAD Mars HH:MM:SS	Nearest REMS UTC est. 2019	REMS LMST HH:MM:SS	UTC gap Earth s	Clock delta Mars m:s	RAD B / E microGy/h	REMS P Pa	REMS air K	REMS flags P / air	Window N
108	09-26 23:41:00	15:06:54	09-26 23:48:18	15:00:46	+437.8	+13:14	11.22455 12.94118	813.55	241.45	10 111XX0011	0
110	09-26 23:58:06	15:23:32	09-26 23:53:25	15:05:45	-281.2	+13:14	11.96270 12.96641	813.27	232.48	11 111XX0001	0
112*	09-27 00:15:07	15:40:06	09-26 23:53:25	15:05:45	-1302.2	+13:14	7.96667 8.88208	813.27	232.48	11 111XX0001	0
114	09-27 00:20:46	15:45:36	09-26 23:53:25	15:05:45	-1641.2	+13:14	11.43665 13.10219	813.27	232.48	11 111XX0001	24
116	09-27 00:37:51	16:02:14	09-27 00:50:00	16:00:49	+728.8	+13:15	12.61789 13.23484	811.58	230.38	10 111XX0011	220
118	09-27 00:54:57	16:18:52	09-27 00:54:57	16:05:38	-0.2	+13:14	12.19637 13.10694	811.37	229.94	11 111XX0011	0
120	09-27 01:12:03	16:35:31	09-27 00:55:07	16:05:47	-1016.2	+13:15	11.45353 12.92504	811.35	230.22	11 111XX0001	0
122	09-27 01:29:09	16:52:10	09-27 01:51:34	17:00:44	+1344.8	+13:15	11.94272 13.03561	812.45	226.00	10 111XX0001	433
124	09-27 01:46:15	17:08:48	09-27 01:51:34	17:00:44	+318.8	+13:14	11.53618 13.08033	812.45	226.00	10 111XX0001	961
126	09-27 02:03:21	17:25:27	09-27 02:03:21	17:12:12	-0.2	+13:15	12.90095 13.13818	813.15	219.53	11 111XX0011	961
128	09-27 02:20:27	17:42:06	09-27 02:20:27	17:28:51	-0.1	+13:15	11.41196 12.84741	814.17	219.89	11 111XX1111	961
130	09-27 02:37:33	17:58:44	09-27 02:37:33	17:45:29	-0.1	+13:15	11.03682 13.03606	815.61	224.89	11 111XX1111	961
132	09-27 02:54:39	18:15:23	09-27 02:54:39	18:02:08	-0.1	+13:15	11.94182 12.86579	817.92	209.83	11 111XX1101	961
134	09-27 03:11:45	18:32:02	09-27 03:11:45	18:18:46	-0.1	+13:16	11.52560 12.94798	821.05	212.08	11 111XX1111	961
136	09-27 03:28:51	18:48:40	09-27 03:28:51	18:35:25	-0.1	+13:15	11.48299 12.82581	824.84	212.29	11 111XX1101	961
138	09-27 03:45:57	19:05:19	09-27 03:45:57	18:52:03	-0.1	+13:16	11.63019 12.86687	828.45	211.80	11 111XX1111	25
140	09-27 04:03:03	19:21:58	09-27 03:59:59	19:05:43	-184.1	+13:16	10.97275 12.92110	830.85	213.57	11 111XX1111	0
142	09-27 04:20:09	19:38:36	09-27 03:59:59	19:05:43	-1210.1	+13:15	11.92671 13.02903	830.85	213.57	11 111XX1111	0

* RAD QC = 00000001; otherwise 00000000. [R] = inferred rollover to sol 2539 (original header: 2538). REMS sol = 2538. -- = missing (-999).

P / air = raw sensor confidence flags; zeros can denote instrument state. No maximum UTC gap was imposed. Clock delta is evaluated at RAD UTC.

Closest-time comparison | 5 of 5

RAD observations 144 to 174 | Sol 2538 | REMS times rounded to nearest second for display

RAD obs.	RAD UTC 2019	RAD Mars HH:MM:SS	Nearest REMS UTC est. 2019	REMS LMST HH:MM:SS	UTC gap Earth s	Clock delta Mars m:s	RAD B / E microGy/h	REMS P Pa	REMS air K	REMS flags P / air	Window N
144	09-27 04:37:15	19:55:15	09-27 04:56:50	20:01:03	+1174.9	+13:16	11.98424 13.02498	835.20	209.38	11 011XX1101	308
146	09-27 04:54:21	20:11:54	09-27 04:56:50	20:01:03	+148.9	+13:16	11.83146 12.81048	835.20	209.38	11 011XX1101	0
148	09-27 05:11:27	20:28:32	09-27 05:01:57	20:06:01	-570.0	+13:16	11.57370 12.82501	835.14	209.57	11 011XX1111	0
150	09-27 05:28:33	20:45:11	09-27 05:01:57	20:06:01	-1596.0	+13:16	12.04645 12.85766	835.14	209.57	11 011XX1111	0
152	09-27 05:45:39	21:01:50	09-27 05:58:19	21:00:53	+760.0	+13:17	11.46201 12.75957	834.84	209.48	11 011XX1111	249
154	09-27 06:02:45	21:18:28	09-27 06:02:45	21:05:12	-0.0	+13:16	12.56841 12.84861	834.99	209.39	11 011XX1111	0
156	09-27 06:19:51	21:35:07	09-27 06:03:26	21:05:52	-985.0	+13:17	11.41146 12.83871	834.92	209.75	11 011XX1111	0
158	09-27 06:36:58	21:51:46	09-27 07:00:07	22:01:02	+1389.0	+13:16	10.77990 12.73854	840.14	203.91	11 011XX1101	308
160	09-27 06:54:04	22:08:24	09-27 07:00:07	22:01:02	+363.0	+13:16	12.04729 12.67995	840.14	203.91	11 011XX1101	0
162	09-27 07:11:10	22:25:03	09-27 07:05:14	22:06:01	-356.0	+13:16	11.92190 12.59715	841.14	203.91	11 011XX1101	0
164	09-27 07:28:16	22:41:42	09-27 07:05:14	22:06:01	-1382.0	+13:16	12.42626 12.55829	841.14	203.91	11 011XX1101	0
166	09-27 07:45:22	22:58:20	09-27 08:02:00	23:01:15	+998.1	+13:16	11.72078 12.54652	850.12	202.54	11 011XX1101	308
168	09-27 08:02:28	23:14:59	09-27 08:02:28	23:01:43	+0.1	+13:16	12.66421 12.67001	850.30	202.92	11 011XX1111	0
170	09-27 08:19:34	23:31:38	09-27 08:07:07	23:06:14	-746.9	+13:17	11.45417 12.61814	850.97	203.56	11 011XX1111	0
172	09-27 08:36:40	23:48:16	09-27 08:07:07	23:06:14	-1772.9	+13:16	11.45655 12.87870	850.97	203.56	11 011XX1111	0
174	09-27 08:53:46	0:04:55 [R]	09-27 08:07:07	23:06:14	-2798.9	+13:17	11.79868 12.82828	850.97	203.56	11 011XX1111	0

* RAD QC = 00000001; otherwise 00000000. [R] = inferred rollover to sol 2539 (original header: 2538). REMS sol = 2538. -- = missing (-999).

P / air = raw sensor confidence flags; zeros can denote instrument state. No maximum UTC gap was imposed. Clock delta is evaluated at RAD UTC.